An Overview of Machine Learning Techniques Used in Real-Time Water Quality Index Measurement

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Abstract — **Water quality index (WQI) has been a universally recognized metric that quantifies water quality, calculated traditionally through analysis of physicochemical characteristics as well as using empirical methods. These conventional approaches can be time-intensive, costly, and may not effectively capture complex relationships among variables. The implementation of machine learning (ML) techniques for more efficient and accurate WQI measurement is the main focus of this paper. Models like Random Forest, Gradient Boosting and so on, required historical data in order to predict WQI dynamically, introducing automation and improving response times. Unique innovations, such as ecological data and hybrid systems, that put together ML and rule-based systems, would improve the accuracy and future development. Results gathered from various experimental data, have demonstrated that ML-based prototypes outperform traditional methods in predictive adaptability, laying foundations for real-time water monitoring solutions.**

Keywords - water quality, machine learning, measurement, real-time.

I. INTRODUCTION

The key component of ensuring the health and safety of ecosystems and the human population is monitoring the world's water resources. WQI is a comprehensive instrument that aggregates several physical, chemical, and biological data within a single number, making the assessment of water quality easier, which aids in determining the levels of pollution, directing water management procedures, and having an influence over legislative choices. Such a strategy includes disciplines with integrated knowledge from environmental science, data science, engineering, and policymaking, necessitates for effective WQI monitoring. Solutions for managing water quality that are more thorough and efficient could result in collaboration between these various fields [1].

Machine Learning (ML) has emerged as an effective technique for environmental monitoring, providing novel answers to conventional problems. ML algorithms can examine massive datasets, detect patterns, and produce very precise predictions. In terms of water quality, models may automate the evaluation process, decreasing the need for considerable human samples and delivering real-time information.

The objective of this paper is to analyze the effectiveness of traditional methods, such as using both versions of the Oregon Water Quality Index (OWQI) [2][3], one of which is with an unweighted harmonic square mean equation and the second, a weighted arithmetic mean function, as well as employing ML methods in WQI analysis.

The main purpose of this paper is to highlight the advantages and limitations of using ML in such an assessment, comparing traditional WQI methods with several ML ones. The paper will also review the historical development of WQI, integration of various techniques, and ensuring the potential improvements that could be offered by ML in relation to efficiency, precision, and cost-effectiveness [4].

II. LITERATURE REVIEW

The first one to establish the notion of WQI was Horton. In 1965, several versions have subsequently been created and enhanced. These models usually require important water quality characteristics, converting raw data to a common scale, weighing each parameter, and aggregating sub-index results. The National Sanitation Foundation WQI (NSFWQI), the Canadian Council of Ministers of the Environment WQI (CCMEWQI), and the Weighted Arithmetic Water Quality Index (WAWQI) are three of the most often utilized WQI models. Many of these models were established in line with specified criteria, and they differ from one another mostly due to the methods utilized in their creation.

For example, the Malaysian WQI (MWQI), developed around 2007, uses six factors: Dissolved Oxygen, Biochemical Oxygen Demand, Chemical Oxygen Demand (COD), Ammonia Nitrogen, suspended solids, and pH. In contrast, the Keya River WQI developed in Taiwan uses thirteen variables: Fecal coliforms, DO, ammonia nitrogen, BOD, suspended solids, turbidity, temperature, pH, toxicity, cadmium (Cd), lead (Pb), copper (Cu), and zinc (Zn). The purpose of all is the same, the measurement of WQI on top of answering to specific requirements and standards by local and/or federal laws. Various areas have circumstances that greatly affect the quality of the water.

The incorporation of the Internet of Things (IoT) in water quality surveillance has further substantially improved the estimation of the WQI [5]. Systems based on IoT use sensors that are interconnected to continually monitor many different variables. These sensors transfer data wirelessly into a central

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framework, allowing real-time analysis and computation of the WQI. The synergy between the IoT and ML not only improves the precision and efficiency of WQI measuring but it also allows for proactive water quality management, opening the potential towards smarter environmental monitoring systems.

III. MACHINE LEARNING TECHNIQUES

The usage of ML in WQI analysis has revolutionized environmental monitoring. ML techniques can interpret massive sets of data, identify patterns, and the production of precise forecasts by putting complex algorithms into practice, thus increasing the accuracy and dependability of assessments of water quality. The next part provides a comprehensive review of machine learning models used in WQI calculation.

Decision Trees (DT) are commonly used in regression and classification applications. They arrange the data into subgroups that correspond to the values in the input characteristics, generating a decision tree-like model. However, they are prone to excessive fitting, especially with complex datasets, and are less effective with huge datasets.

Random Forest (RF) is a learning technique that enhances prediction accuracy by generating many decision trees and pooling their results. Despite its benefits, Random Forest might be more computationally intensive and difficult to understand compared to singular decision trees.

For both classification and regression, **Support Vector Machines (SVMs)** are usually used. They work by identifying the most effective hyperplane for classifying data. SVMs are well known for handling complex, nonlinear data with remarkable accuracy and effectiveness. These procedures are quite computationally intensive and require exact parameter tuning to achieve optimal efficiency.

Artificial neural networks (ANNs) are modeled after the human cognitive system and consist of linked nodes, brain cells that process data. ANNs are extremely adept at managing complicated non-linear connections in data, resulting in high prediction accuracy. However, they require large datasets and are computationally expensive, making them less interpretable than simpler models.

Gradient boosting machines (GBMs) are powerful ensemble learning approaches for enhancing the accuracy of prediction. Sequentially, this technique is well-known for its high accuracy and long-lasting performance, but it is computationally intensive and, if not well-regulated, can result in overfitting.

Convolutional Neural Networks (CNN), which initially emerged for image and video recognition, have since been utilized for an array of programs, including WQI prediction. They require massive databases that are computationally intensive.

Long Short-Term Memory (LSTM) networks are a similar type of recurrent neural network that was designed specifically for recognizing long-term dependence in sequential input. LSTMs are highly accurate regarding time-series predictions and perform well with sequential data.

Those represent some of the most common machine learning methods utilized during WQI computations, and each is adapted to a certain sort of data and goal.

According to the following research presented in [6], the main objective was to explore the application of ten machine learning models to estimate the WQI for the Southern Bug River in Ukraine. The models that were evaluated were the Gaussian Process (GP) and XGBoost for their effectiveness in predicting WQI from various water quality parameters. The former was found to outperform the others, displaying robust performance and reliability, and thus, has proven its suitability for WQI estimation.

In [7], feature selection approaches were integrated with models to forecast WQI for the An Kim Hai irrigation system. There were identified four critical factors (coliform, DO, turbidity, and TSS) and an RF model was created that was very accurate. The study's strength comes in lowering the number of input factors, thus improving monitoring efficiency.

In [8], the result of using supervised ML algorithms to predict water quality using parameters like temperature, turbidity, pH, and TDS has been presented. Gradient boosting and polynomial regression models showed the best performance. Gradient Boosting effectively improves accuracy through iterative boosting and the handling of nonlinear relationships, while Polynomial Regression models nonlinear patterns found in the data.

For the research presented in [9], applied ML models, Gradient Boosting in particular, were used to predict WQI for an industrially polluted stream in Pakistan. The study used 19 input variables and found that the Gradient Boost model was the most effective. Perfect for WQI prediction in polluted environments due to its robustness in handling interactions of specific complexity between multiple variables and its ability to improve accuracy through boosting.

In [10], the application of techniques, such as LSTM, to forecast WQI for the Kelang River is described. The LSTM system performed the best, achieving results with great precision. It can capture temporal dependencies in sequential data, which makes it useful for time-series analysis and suitable for WQI prediction.

In [11], an IoT-based WQI monitoring system for aquaculture using a fuzzy inference system and multi-gene genetic programming was presented. The system demonstrated good prediction accuracy, but it is unique to Nile tilapia, making it perfect for WQI prediction in aquaculture due to its capacity to build nonlinear equations, its capability to model complex relationships, and provide precise real-time monitoring.

In [12], the usage of grid search to optimize ML models for predicting WQI and WQC is examined. The Gradient Boosting model produced decent classification accuracy, while the Multilayer Perception regressor was particularly effective at predicting WQI. Future research should use recurrent neural networks for time-series prediction, according to the paper.

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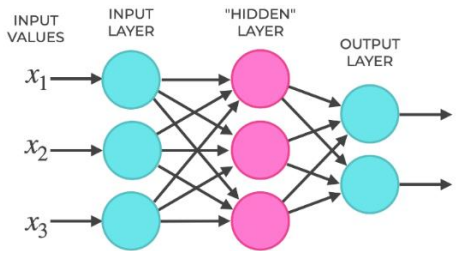


Figure 1: Multilayer Perception model

This case study [13] evaluates the J48, Naïve Bayes, and MLP algorithms for water quality classification. The MLP algorithm has the best accuracy. It is effective for water quality classification because it can simulate complicated, nonlinear interactions between water quality measures while maintaining high accuracy. This study did not investigate IoT integration or other methodologies, for example, deep learning, which might be areas of future research.

According to the study provided in [14], multiple ML algorithms were employed to classify water quality in India. The Multinomial Logistic Regression (MLR) model produced the best results. It is suitable for WQI classification because it can handle many classes, provide probabilistic predictions, and control feature significance using regression coefficients.

The research reported in [15] uses ML models to predict and classify river water quality indexes. The MLP regressor and classifier fared the best, therefore, it was chosen for its capacity to model nonlinear relationships, produce accurate river water quality forecasts, and successfully handle high-dimensional data.

In [16], Sentinel-2 MSI data and ML models are used to monitor water quality in Ebinur Lake. The Partial Least Squares Regression-Support Vector Machines model performed best. It is designed for WQI prediction using remote sensing data due to its ability to handle high-dimensional spectral data, model nonlinear relationships, and provide robust predictions. The study's limitation is the geographic scope, and future research could expand to other watersheds.

In [17], the ANN, DT, and SVM model algorithms are compared for the classification of water quality in the Langat River Basin. The SVM model outperformed others for the given task due to its suitability to handle high-dimensional data, nonlinear decision limits, and provide accurate forecasts.

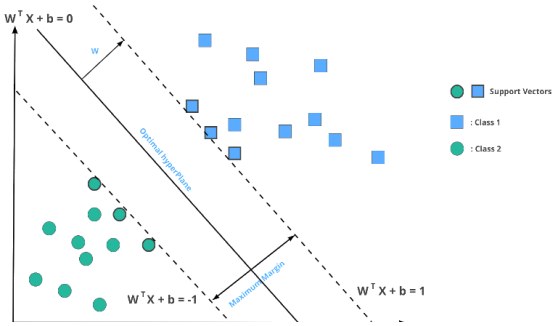


Figure 2: Support Vector Machine algorithm

In [18], eight regression algorithms are evaluated. Linear regression and ridge regression performed best. They are qualified for WQI prediction as they provide interpretable models, handle multicollinearity through regularization, and extend elementary parameter choice.

According to [19], ML models are used to forecast water quality in the Tumkur district, India. Support Vector Machines (SVM) and Linear Regression (LR) outperformed other models. SVMs can capture complex patterns and handle nonlinear relationships, while LR provides a simple approach for WQI forecasting.

Table 1. Table overview of the reviewed literature

Reference	Models Used	Key Findings
[6], [8], [9]	Gaussian Process, XGBoost	GP outperformed others.
[7]	Random Forest	Accurate with four critical factors.
[10]	Long Short-Term Memory	Best for time-series analysis.
[11]	Fuzzy Inference, Genetic Programming	Accurate for aquaculture.
[12]	Gradient Boosting, Multilayer Perception	Effective for prediction and classification.
[13], [15]	Multilayer Perception	Best accuracy for classification.
[14]	Multinomial Logistic Regression	Best for WQI classification.
[16], [17], [18], [19]	Partial Least Squares Regression-Support Vector Machines, Linear Regression, and Ridge Regression	Qualified for WQI prediction.

RF, Gradient Boosting (GB), and SVM were clearly the most popular models because to their resilience and accuracy in forecasting the WQI. RF and GB, as both are approaches, regularly beat standalone methods like LR and DT by lowering variance and increasing the accuracy of prediction by combining numerous weak learners. In occurrences where interpretability is critical, such as regulatory compliance or public health alerts, simpler models, such as LR or DT, may be preferable, despite their lower accuracy.

IV. METHODOLOGY

Sources of Data and Preprocessing: A wide range of pre-existing datasets, with a variety of variables, were used to generate the data for the current study. These datasets incorporated measurements of several parameters, as well as Biological Oxygen Demand (BOD), Total dissolved solids (TDS), turbidity, pH, and Dissolved oxygen (DO). Data preparation included several key processes to ensure data accuracy and reliability. This includes methods for value input that are not present, data normalization, and feature selection to determine the crucial variables for predicting the WQI.

Hyperparameters and ML Algorithms: This means the implementation of ML algorithms to forecast the WQI. These included RF, GB, SVM, MLP, and LSTM networks. Each algorithm should be fine-tuned using hyperparameter optimization techniques.

Model Evaluation along with Comparison: For the main purpose of assessing and contrasting the performance of the ML

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models, a series of procedures can be implemented. Cross-validation methods such as k-fold cross-validation were employed to ensure the robustness of the model's performance metrics, which used mean absolute error (MAE), root mean squared error (RMSE), R-squared (R^2), and precision regarding tasks about classification.

V. CONCLUSION

Using a huge set of water quality indicators, the adoption of several ML models is investigated to forecast the WQI in this work. The current investigation showed how well ML algorithms, RF, and GB, provide precise and trustworthy projections. These models were applicable for convoluted environmental datasets because they have proven their worth regarding the demonstration of exceptional proficiency in dealing with high-dimensional data and minimization of variation.

Feature selection, hyperparameter tuning, and extensive data preliminary treatment to ensure the models' quality and resilience were included in the study. Cross-validation techniques and performance metrics such as Mean Absolute Error and Root Mean Squared Error are usually used to rate the model's particular performance in detail.

Based on the analysis, the best-suited machine learning models for Water Quality Index prediction are Random Forest, Gradient Boosting, Support Vector Machines, Artificial Neural Networks, Multilayer Perception Regressor, and Long Short-Term Memory Networks.

There are significant implications when ML is applied to real-time applications and comprehensive WQI monitoring. ML models can predict water quality fast and accurately, enabling proactive management and timely action. However, putting these models into practice requires careful consideration of computing resources and data integration from multiple sources.

In conclusion, the improvement of water quality control and monitoring with ML for WQI forecasting is straightforward. By addressing the identified limitations and utilizing current advances in ML and the IoT, more dependable, scalable, and easily accessible models that support sustainable management of water resources and protect public health could be developed.

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